



ACM

**ACM Code of Ethics and Professional Conduct,
ACM/IEEE Software Engineering Code of Ethics and
Professional Practices**



Introduction

- What is the ACM (Association for Computing Machinery) Code of Ethics?
 - - A ***contract among computing professionals*** guiding ethical conduct.
 - - Emphasizes ***responsibility to society, human well-being, and the environment***.
 - - First established in 1992, updated in 2018 to reflect modern challenges.
 - Why is it Important?
 - - Ensures computing is used for public good.
 - - Protects sensitive information, minimizes harm, and promotes trust
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Professional Codes of Ethics

- Many professions have established professional societies, which have adopted codes of conduct.
 - ❑ AMA (American Medical Association)
 - ❑ ABA (American Bar Association).
- Two computing professional societies:
 - ❑ The Association for Computing Machinery (ACM)
 - ❑ The Institute for Electrical and Electronics Engineers – Computer Society (IEEE-CS)

Purpose of Professional Codes

- Professional codes of ethics are often designed to motivate members of an association to behave in certain ways.
 - Four primary functions of codes are to:
 - *inspire*
 - *guide*
 - *educate*
 - *discipline* the members.
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ACM Code of Ethics and Professional Conduct

<https://www.acm.org/code-of-ethics>

ACM (Association for Computing Machinery) Code

- A Code of Professional Conduct (passed by the ACM Council in 1973) specified that:
 - An ACM member, whenever dealing with data concerning individuals, shall always consider the principle of the individuals' privacy and seek to:
 - **Minimize the data collected**
 - **Limit authorized access to the data**
 - **Provide proper security for the data**
 - **Determine the required retention period of the data**
 - **Ensure proper disposal of the data**
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- When the code was revised in 1992, these edicts were dropped and replaced with a General Moral Imperative specifying that an ACM member will
 - **“Respect the privacy of others.”**
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Computer professionals Responsibility

- Computing professionals' actions should reflect upon the wider impacts of their work, consistently supporting the **public good**.
- The **Code is designed** to guide the ethical conduct of all computing professionals.
- Additionally, the Code serves as a basis for remediation when violations occur.
- The Code includes principles formulated as statements of responsibility.

ACM code outline:

- **Section 1** outlines fundamental ethical principles that form the basis for the remainder of the Code.
 - **Section 2** addresses additional, more specific considerations of professional responsibility.
 - **Section 3** guides individuals who have a leadership role.
 - Principles involving compliance with the Code are given in **Section 4**.
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ACM Code of Ethics and Conduct (section 1)

- **1.1 Contribute to society and human well-being.**
 - **1.2 Avoid harm to others.**
 - **1.3 Be honest and trustworthy.**
 - **1.4 Be fair and take action not to discriminate.**
 - **1.5 Respect the work required to produce new ideas, inventions, creative works, and computing artifacts.**
 - **1.6 Respect privacy.**
 - **1.7 Honor confidentiality.**
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ACM Section 2 MORE SPECIFIC PROFESSIONAL RESPONSIBILITIES.

ACM computing professional should

- **2.1 Strive to achieve high quality in both the processes and products of professional work.**
 - **2.2 Maintain high standards of professional competence, conduct, and ethical practice.**
 - **2.3 Know and respect existing rules pertaining to professional work.**
 - **2.4 Accept and provide appropriate professional review.**
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- 2.5 Give **comprehensive and thorough evaluations of computer systems** and their impacts, including analysis of possible risks.
 - 2.6 Perform work only in areas of competence/expertise.
 - 2.7 Foster **public awareness and understanding of computing**, related technologies, and their consequences.
 - 2.8 Access computing and communication resources only when authorized or when compelled by the public good.
 - 2.9 Design and implement systems that are robustly and useably secure.
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ACM Section 3 ORGANIZATIONAL LEADERSHIP IMPERATIVES.

- **This section draws extensively from the draft of IFIP (*International Federation for Information Processing*) Code of Ethics.**
- **The ethical obligations of organizations tend to be neglected, *perhaps because these codes are written from the perspective of the individual member.***
- ***This dilemma is addressed by stating these **imperatives from the perspective of the organizational leader.*****
- **In this context "leader" is viewed as any organizational member who has leadership or educational responsibilities.**

ACM member and an organizational leader should

- **3.1 Ensure that the **public good** is the central concern during all professional computing work.**
- **3.2 Articulate, encourage acceptance of, and evaluate **fulfillment of social responsibilities** by members of the organization or group.**
- **3.3 Manage personnel and resources to enhance the quality of working life.**

- **3.4 Articulate, apply, and support policies and processes** that reflect the principles of the Code.
- **3.5 Create opportunities for members of the organization or group to grow as professionals.**
- **3.6 Use care when modifying or retiring systems.**
- **3.7 Recognize and take special care of systems that become integrated into the infrastructure of society.**

ACM; Section 4 COMPLIANCE WITH THE CODE.

As ACM member should

- **4.1 Uphold and promote the principles of this Code.**
- **4.2 Treat violations of this code as inconsistent with membership in the ACM.**

The Software Engineering Code of Ethics and Professional Practice

- The ACM and IEEE-CS (Institute for Electronics and Electrical Engineering Computer Society)
- **Software Engineering Code of Ethics** and Professional Practice as recommended by the ACM/IEEE-CS Joint Task Force on Software Engineering Ethics and Professional Practices and jointly approved by the **ACM and the IEEE-CS** as the standard for teaching and practicing software engineering.
- <https://ethics.acm.org/code-of-ethics/software-engineering-code/>

Eight Principles Software engineers should follow:

- Software engineers shall commit themselves to making the analysis, specification, design, development, testing and maintenance of software a beneficial and respected profession.
 - In accordance with their commitment to the health, safety and welfare of the public, **software engineers shall adhere to the following Eight Principles:**
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- **1. PUBLIC** – Software engineers shall act consistently with the public interest.
 - **2. CLIENT AND EMPLOYER** – Software engineers shall act in a manner that is in the best interests of their client and employer consistent with the public interest.
 - **3. PRODUCT** – Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.
 - **4. JUDGMENT** – Software engineers shall maintain integrity and independence in their professional judgment.
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- **5. MANAGEMENT** – Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.
- **6. PROFESSION** – Software engineers shall advance the integrity and reputation of the profession consistent with the public interest.
- **7. COLLEAGUES** – Software engineers shall be fair to and supportive of their colleagues.
- **8. SELF** – Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Conclusion

Key Takeaways:

- - The ACM Code of Ethics promotes integrity, public good, and professional responsibility in computing.
 - - Ethical decision-making requires ongoing reflection and commitment to the Code.
 - - Professionals must be aware of the societal impact of their work and strive for excellence in both technical and ethical conduct.
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Whistleblowing:

- **Whistleblowing** refers to the act of exposing information or activity within an organization that is deemed illegal, unethical, or not correct. A whistleblower can be an employee, contractor, or member of an organization who reports misconduct to people or entities that have the power to take corrective action.
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Why Do Individuals Choose to Engage in Whistleblowing?

- People blow the whistle for several reasons, including:

1. **Moral or Ethical Concerns**

They feel a personal or professional responsibility to report wrongdoing, especially if it harms others or violates societal values.

2. **Legal Obligation**

In some sectors, such as finance, healthcare, or public service, individuals may be legally required to report misconduct.

3. **Protecting the Public Interest**

Whistleblowers often act to prevent harm to the public, such as in cases of environmental violations, health and safety hazards, or financial fraud.

Why Do Individuals Choose to Engage in Whistleblowing?

- **Loyalty to the Organization**

Some whistleblowers aim to help the organization correct internal problems before they escalate or cause external damage.

- **Retaliation or Personal Grievances**

In some cases, whistleblowing may be driven by interpersonal conflicts or as a response to unfair treatment within the organization.

- **External Pressure or Encouragement**

Government agencies, media, or watchdog organizations sometimes provide channels and protection that encourage people to come forward.
